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STAGE: STEREOTYPICAL TRUST ASSESSMENT THROUGH GRAPH EXTRACTION

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Bootstrapping trust assessment where there is little or no evidence regarding a subject is a significant challenge for existing trust and reputation systems. When direct or indirect evidence is absent, existing approaches usually assume that all agents are equally trustworthy. This naive assumption makes existing approaches vulnerable to attacks such as *Sybil* and *whitewashing*. Inspired by real-life scenarios, we argue that malicious agents may share some common patterns or complex features in their descriptions. If such patterns or features can be detected, they can be exploited to bootstrap trust assessments. Based on this idea, we propose the use of frequent subgraph mining and state-of-the-art knowledge representation formalisms to estimate *a priori trust* for agents. Our approach first discovers significant patterns that may be used to characterize trustworthy and untrustworthy agents. Then, these patterns are used as features to train a regression model to estimate the trustworthiness of agents. Last, a priori trust for unknown agents (e.g., newcomers) is estimated using the discovered features based on the trained model. Through empirical evaluation, we show that the proposed approach significantly outperforms well-known trust approaches if trustworthiness of agents is correlated with patterns in their descriptions or social networks. Furthermore, we show that the proposed approach performs at least as good as the existing approaches if such correlations do not exist.

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1. INTRODUCTION

In open systems like the Web, there is no central authority that monitors interacting agents and guarantees that every agent in the system behaves as expected. For instance, a seller in an e-marketplace may list a product at a cheaper price but may fail to deliver the advertised product or anything at all. This uncertainty derives the need for agents to evaluate others to inform the selection of interaction partners. While the word "trust" may have different definitions in different domains, we define it here pragmatically as the degree of belief or subjective probability, with which a *truster* (e.g., buyer) believes that a *trustee* (e.g., seller) will perform as expected when relied upon (Jøsang, Ismail, and Boyd 2007).

Existing statistical approaches for trust generally depend on an interaction history to compute the trust of an agent x in another agent y with respect to some task t. The interaction history contains the interactions between x and y (i.e., direct evidence) and their outcomes (e.g., *success* and *failure*). If the number of the direct interactions is insufficient to compute a trust value with sufficient confidence, reputational evidence (i.e., indirect evidence) may be used. Outcomes of prior interactions between other agents and y serve as reputational evidence. Statistical trust approaches such as the beta reputation system (BRS) (Jøsang and Ismail 2002), TRAVOS (Teacy et al. 2006), BLADE (Regan, Poupart, and Cohen 2006), and HABIT (Teacy et al. 2012) use both direct and indirect evidence to compute such trust values. Here, a trust assessment represents the subjective probability that a

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future interaction with y would have a desirable outcome. As the quantity and quality of evidence increase, the accuracy of the estimated trustworthiness increases, and the number of unsatisfactory interactions is minimized.

Although the existing approaches can accurately model trust, they require repetitive interactions between agents. This requirement leads to two related problems: *bootstrapping* and *whitewashing*. The bootstrapping problem arises when a new agent joins to the system. In this case, trust cannot be computed based on direct or reputational evidence. To deal with the bootstrapping problem, existing approaches assume an a priori trust value for the newcomers. For instance, in BRS and TRAVOS, the a priori trust is 0.5, which means that positive and negative outcomes are equally likely. If the a priori trust is high, agents with bad reputations have an incentive to whitewash; i.e., re-enter the society with a new identity as if there were a newcomer. On the other hand, if the a priori trust is too low, genuine newcomers are likely to have fewer opportunities to interact with others and build a good reputation.

Recently, bootstrapping trust in the absence of evidence (i.e., interaction history) has drawn the attention of the research community. Liu et al. (2009) and Burnett, Norman, and Sycara (2010, 2013) have proposed solutions based on stereotyping through machine learning. They suggest that agents learn a mapping between known agents' observable features and their trustworthiness. Once this mapping is learned, it is used to compute the a priori trust for unknown agents (the newcomers) based on their observable features. Hence, instead of using a single a priori trust for all, these approaches may determine different a priori trust values for different types of agents.

Stereotyping exploits a possible mapping between observable features of individuals and their behavior and is shown to be useful in both human (Hilton and von Hippel 1996) and agent (Liu et al. 2009; Burnett et al. 2010, 2013) societies to bootstrap trust in the absence of evidence. However, existing approaches for stereotyping consider only simple attributes (e.g., age and location) of an agent as its features. Therefore, these approaches may learn stereotypes capturing the mappings between trust and simple attributes (e.g., "agents from country X are less trustworthy"), but they fail to discover the connection between trustworthiness and complex (and even dynamic) features (e.g., "agents with specific patterns of relationships are less trustworthy").

Malicious agents may adopt certain behavioral patterns to achieve their goals. These patterns may determine their choices and lead to the emergence of certain motifs in their relationships with other entities. Even if an agent whitewashes its identity and changes or fakes some of its observable attributes, the same motifs may be observed as long as the agent does not change its behavior. For example, a malicious seller may change its identity and advertise a completely new profile whenever its reputation decreases. Although the *name*, *location*, *e-mail*, and *Web site* in the new profile are different from previous ones, all these e-mail addresses and Web sites could be hosted by the same or similar service providers (e.g., free hosting services). If malicious sellers have a tendency to bear the same or similar pattern in their profiles, we may build a stereotype such as "sellers using free hosting services are less trustworthy." This stereotype is not based on a simple attribute of the seller (e.g., a specific e-mail address) but an example of a complex feature discovered about the malicious sellers.

This article proposes to discover and exploit complex features (i.e., patterns) of agents to estimate a priori trust for them. Knowledge about each known agent is represented in detail as a graph based on an ontology. Then, these graphs are mined to discover two groups of patterns that frequently appear in trustworthy and untrustworthy agents, respectively. Last, the discovered patterns are used as features to train a regression model to estimate the a priori trust for the unknown agents. Through extensive simulations, we have

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showed that the proposed approach significantly outperforms existing trust approaches in challenging settings.

The rest of the article is organized as follows. In Section 2, we introduce Subjective Logic (SL) for modeling trust. Sections 3 and 4 propose a graph-based representation to describe agents and their relationships based on an ontology, and the techniques we propose for discovering significant patterns in these graphs and estimating a priori trust based on patterns discovered. In Section 5, we describe how the proposed approach can be used in combination with direct and indirect evidence, and we empirically demonstrate the effectiveness of our approach in Section 6. We set this research in the context of the relevant literature and highlight avenues for future research in Section 7. Finally, we conclude with an overview of our contributions in Section 8.

2. MODELING TRUST USING SUBJECTIVE LOGIC

Several approaches have been proposed to model trust in the literature (Jøsang et al. 2007). A number of these approaches are based on *SL*, which is a belief calculus that allows agents to express opinions as degrees of belief, disbelief, and uncertainty about propositions. Let ρ be a proposition such as "agent y is trustworthy in context c." The binomial opinion about the truth of a proposition ρ is represented as the tuple (b, d, u), where b is the belief that ρ is true, d is the belief that ρ is false, and u is the uncertainty, such that $b+d+u = 1.0$ and $b, d, u \in [0, 1]$.

Posteriori probabilities of binary events can be represented as beta distributions (Jøsang 2011). The beta family of probability density functions is a family of functions with two parameters: α and β. The beta probability distribution $f(p|\alpha, \beta)$ is formalized in equation (1) using the gamma function Γ, where $0 \leq p \leq 1, 0 < \alpha$, and $0 < \beta$. The probability expectation value of the beta distribution is calculated as $E(p) = \alpha/(\alpha + \beta)$.

$$f(p|\alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1}(1 - p)^{\beta-1} . \tag{1}$$

Let us assume that a binary event has two possible outcomes $\{x, \bar{x}\}$ and the numbers of observed outcome for x and $\bar{x}$ be r and s, respectively. Then, we can formalize a beta probability density distribution of observing the outcome x in the future using parameters $\alpha = r + 1$ and $\beta = s + 1$.

The SL uses beta probability density distributions to represent binomial opinions. That is, in SL, the binary opinion of agent x about the proposition ρ is equivalent to a beta distribution. Opinions are formed on the basis of positive and negative evidence observed in the past. Let r and s be the number of positive and negative evidences (i.e., equivalently weighted pieces of evidence) about the proposition ρ, respectively. Then, b, d, and u are computed based on equation (2)

$$b = \frac{r}{r + s + 2}, \; d = \frac{s}{r + s + 2}, \; u = \frac{2}{r + s + 2} . \tag{2}$$

There is a bijection between the opinion (b, d, u) and associated evidence (r, s). Given an opinion, positive evidence and negative evidence are computed as in equation (3).

$$r = \frac{2 \times b}{u} \text{ and } s = \frac{2 \times d}{u}. \tag{3}$$

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Let $b(w)$, $d(w)$, and $u(w)$ be functions returning the belief, disbelief, and uncertainty in an opinion w. Similarly, let $r(w)$ and $s(w)$ be functions that map w to positive and negative evidence (i.e., r and s), respectively, based on equation (3). Each opinion is associated with a *base rate* $a \in [0, 1]$, which is the a priori probability in the absence of evidence. Using the base rate, the opinion's probability expectation value is computed using equation (4). The computed expectation value for the opinion about ρ can be used by x as the trustworthiness of y in context c, which we refer to as $t^x_{y:c}$. For clarity, in this article, we assume that trust is computed by the same truster agent x in the same context c, unless otherwise is stated. Therefore, we abbreviate $t^x_{y:c}$ to t_y.

$$E(w, a) = b(w) + a \times u(w) = \frac{r(w) + a \times 2}{r(w) + s(w) + 2}\,. \tag{4}$$

The base rate parameter a represents the a priori degree of trust x has about y in context c before any evidence has been acquired. The default value of a is typically chosen to be 0.5 (Jøsang 2011), which means that before any positive or negative evidence has been acquired, both outcomes are considered equally likely. As x accumulates more evidence to evaluate the trustworthiness of y, the uncertainty u and the influence of a in the expectation value calculations decrease.

Modeling trust based on subjective opinions allows us to exploit secondary probabilities. A binary opinion calculated based on evidence (r, s) actually corresponds to a beta probability distribution function (Jøsang et al. 2007) with parameters $(r + 1, s + 1)$. Figure 1 shows how the beta distributions are shaped based on the amount of evidence. The expectation value of the distribution is computed as in equation (4).

In this article, to take into account the dynamicity in the environment (e.g., behavior change), only the most recent pieces of evidence are considered. That is, time is divided into windows by the agent who will evaluate trust. Then, positive and negative evidence only from the most recent time window is used to compute trust. The optimum selection of a time window is outside the scope of this article, and a detailed discussion about it can be found elsewhere (Zhang and Cohen 2006; Şensoy et al. 2009).

3. KNOWLEDGE REPRESENTATION

To describe agents and their relationships semantically and flexibly, we propose to use an ontology (Antoniou and van Harmelen 2004). To demonstrate toy examples in the article,

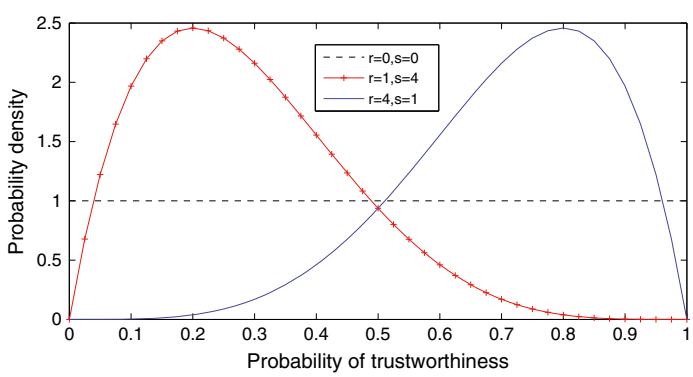

FIGURE 1. Beta probability density functions for different amounts of evidences.

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Figure 2 shows a part of a simple ontology, where arcs, ellipses, and rectangles represent relationships, concepts, and their instances, respectively. The description of an agent is represented using $\langle subject, relation, object \rangle$ triples, such as $\langle john, hasLocation, Spain \rangle$. In these triples, subjects, objects, and properties are terms from the ontology. That is, subjects are instances of concepts (e.g., *john*); objects are literals (e.g., 60), concepts (e.g., *Location*), or their instances (e.g., *Spain*); and relations are data type properties (e.g., *hasAge*) or object properties (e.g., *hasLocation*).

Each known agent is represented as an instance (e.g., an instance of *Seller* concept) in the ontology and described using $\langle subject, relation, object \rangle$ triples. The description of agent y can be semantically represented as a labeled directed graph G_y. The node representing y in G_y is called the terminal node and is connected to other nodes through relationships from the ontology. Nodes in G_y refer to either literals or instances, and edges of G_y correspond to relationships between them. A seller agent *john* may be represented as the graph shown in Figure 3. Given the ontology, this graph describes the seller *john*. However, it is not possible to completely interpret or reason about this description without the ontology, because the terms (i.e., individuals and properties) used in this graph are defined within this ontology. That is why this graph is called *ontology-dependent* graph.

Almost none of the state-of-the-art graph mining tools incorporates ontological knowledge (Donato and Gionis 2010). Hence, most existing tools neglect the semantics of labels in a graph. For instance, for these approaches, *Spain* in the graph of Figure 3 is just a label of a node. However, the ontology of Figure 2 implies that *Spain* is a location in Europe. Therefore, ontology-dependent graphs are not very informative for existing graph mining approaches (Yan and Han 2002). One way of making them more informative as stand-alone graphs is to embed relevant ontological knowledge into them. For this purpose, we use an ontological reasoner (we uses Pellet (Sirin et al. 2007)) to derive all direct and indirect statements (i.e., triples) about the individuals in the graph. Let us note

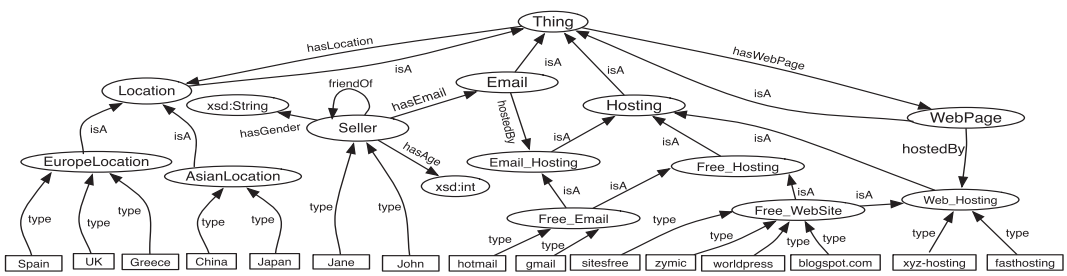

FIGURE 2. A part of a simple ontology to describe sellers, their properties, and relationships.

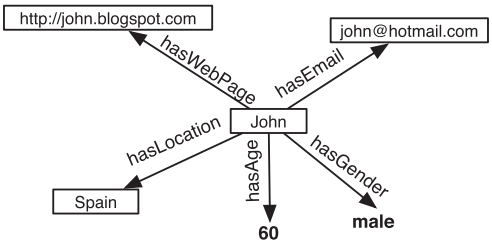

FIGURE 3. Ontology-dependent description of the seller *john*.

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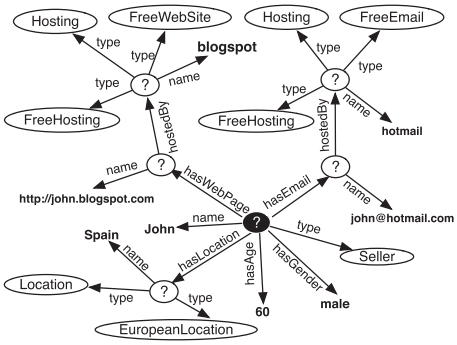

FIGURE 4. Ontology-independent description of the seller *john*.

that these individuals correspond to nodes in the original graph. Then, based on each triple $\langle subject, relation, object \rangle$, we create new nodes and relations if they are missing in the graph. As a result, the graph contains all direct and inferred relevant information at different levels of abstraction. Using individual's names as node labels may hamper frequent pattern mining, because these individuals may not appear frequently in the data set. To avoid this, we replace these labels with "?" and add name (i.e., URI) of the referred individuals as a property of these nodes, i.e., using the *name* data type property. The resulting graph is called *ontology-independent* graph and referred to as g_y, because we do not need an ontology to infer the properties of individuals on the graph; instead, each node referring to an individual bares all direct and inferred attributes and relationships of the individual. Figure 4 shows the ontology-independent version of the graph shown in Figure 3. The terminal node is colored *black* in the graph.

Although we use a simple toy example to illustrate our graph-based representation here, this representation is very expressive and flexible enough to accommodate not only the attributes of the trustee but also all of its relationships with other entities (e.g., *friend of friend* relationships), as demonstrated in Section 5.

4. GRAPH MINING FOR ESTIMATING A PRIORI TRUST

In the previous section, we described how the information about an agent can be represented as a graph, which captures various aspects of the agent such as its attributes and even its relationships with other agents at different levels of abstraction. In this section, we propose to exploit graph mining techniques to discover patterns that may exist more frequently in either trustworthy or untrustworthy agents. For this purpose, we create graph data sets composed of graphs describing agents. Once the frequent patterns have been discovered, we use them as features to estimate a priori trust.

4.1. Determination of Significant Patterns

For each known agent y, an ontology-independent graph g_y is generated to describe the agent. Then, g_y is labeled (i.e., classified) based on the trustworthiness of y, i.e., t_y; the labels are the categories based on the degree of trustworthiness. In this work, for the sake clarity and simplicity, we use only two categories: *trustworthy* and *untrustworthy*. Therefore, in the resulting graph data set, there are two classes of graphs: C_+ and C_-, which represent the graphs generated from *trustworthy* and *untrustworthy* agents, respectively. An agent y is considered trustworthy if $t_y \geq \gamma$ and untrustworthy if $t_y \leq \delta$ where $0 \leq \delta < 0.5 < \gamma \leq 1$. We want to discover two sets of patterns: P_+ and P_-, which are called patterns of *trust* and

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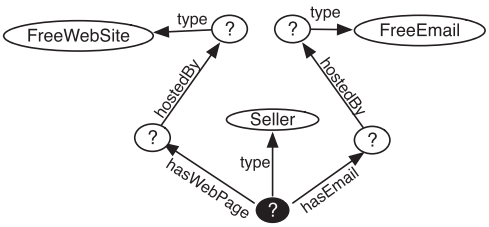

FIGURE 5. A frequent pattern for untrustworthy sellers.

distrust, respectively. A pattern $p \in P_-$ is a subgraph that repeats frequently in the graphs of C_- and rarely in the graphs of C_+. Therefore, p tends to distinguish untrustworthy agents from trustworthy ones. For instance, if most of the known sellers using free e-mail and Web hosting services are untrustworthy, the proposed approach may discover the pattern in Figure 5 as a frequent pattern distinguishing untrustworthy sellers from trustworthy ones. Similarly, P_+ represents the significant patterns that repeat more frequently in the graphs of C_+. A formal definition of the patterns in P_- and P_+ is given in Definition 1 based on the frequencies explained in Definition 2.

Definition 1. Let $\nu \in \{+, -\}$ and $\bar{\nu} \in \{+, -\} \setminus \{\nu\}$. A frequent pattern $p \in P_\nu$ is a pattern such that $f(p, C_\nu) \geq \alpha_\nu > \sigma f\ (p, C_{\bar{\nu}})$, where $f(p, C_\nu)$ and $f\ (p, C_{\bar{\nu}})$ are frequencies of p in classes C_ν and $C_{\bar{\nu}}$, respectively; $0 < \alpha_\nu \leq 1$ is a threshold, and $\sigma \geq 1$ is a coefficient.

Definition 2. Given a pattern p, $f(p, C_\nu)$ is the frequency of the pattern in the graphs of class C_ν and computed by the formula:

$$f(p, C_\nu) = \frac{N_{p:C_\nu}}{|C_\nu|}$$

where $N_{p:C_\nu}$ and $|C_\nu|$ are the number of graphs containing p in class C_ν and size of C_ν, respectively.

Without support constraints (i.e., α_+ and α_-), we have to deal with huge number of patterns. To narrow down the search space, the truster agent determines these constraints. An additional constraint can be the *minimum size* of the frequent patterns, where the size of the pattern is determined by its number of edges. In the literature, a number of *frequent subgraph mining* approaches have been proposed to extract frequent patterns in multi-class graph data sets using support and size constraints (Cook and Holder 2000; Yan and Han 2002; Chakrabarti and Faloutsos 2006).

4.2. Estimating A Priori Trust

As described earlier, the truster agent finds two sets of frequent patterns (P_+ and P_-) using frequent subgraph mining. In this section, we describe how these significant patterns can be used as features to estimate a priori trust for agents. Then, the estimated a priori trust is used as the base rate a in equation (4) while computing the trustworthiness of the agents.

To estimate the a priori trust, the truster can use various machine learning techniques (Alpaydin 2001). In this work, we employ the $M5$ regression tree algorithm (Quinlan 1992) for this purpose. Given a training set and a set of features $\mathcal{F}$, this algorithm learns a mapping between feature values of an agent and its trustworthiness. In this article, each feature $f_j \in$

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$\mathcal{F}$ corresponds to a specific pattern $p_j \in P_+ \cup P_-$, and its value for an agent i indicates if the graph g_i describing the agent contains (i.e., entails) p_j or not.

The training set is prepared using the descriptions of known agents and their trustworthiness. That is, for each known agent i, the training set contains a feature vector $\vec{v_i}$ and trust value t_i. Each field v_{ij} in the feature vector corresponds to the value of feature f_j for the agent i, which is determined by checking entailment of p_j by g_i. That is, $v_{ij} = 0$ if the pattern p_j does not exist in the graph g_i describing i; otherwise, $v_{ij} = 1$. In this way, we transform agents' graphs of various sizes into a vector data set with fixed dimensions. After creating feature vectors for each known agent i, the set of $\left(\vec{v_i}, t_i\right)$ pairs is used as a training set to learn the regression model. The trained regression model is used as a function $R : \vec{v} \rightarrow [0, 1]$ that takes the feature vector $\vec{v}$ of an agent as input and returns an estimate of its trustworthiness. This estimate is not based on any evidence about the target agent but only its features; hence, what is estimated is the a priori trust in the target agent.

To compute trustworthiness of an unknown agent y, first, the feature vector $\vec{v_y}$ is created based on the discovered features. Then, t_y is computed using equation (4) where $R\left(\vec{v_y}\right)$ is used as the base rate a.

The truster may have more interactions over time and learn trustworthiness of others better based on new evidence. Meanwhile, the society may change, and new behavioral patterns may be adopted by malicious agents. Therefore, the truster may periodically use the described approach to discover new features significant for trustworthiness and retrain the regression model based on these new features.

5. COMBINING DIRECT AND INDIRECT EVIDENCE

The trust model we introduced in Section 2 is based on direct evidence. Unfortunately, direct evidence is often limited or may not be available while computing trust. Therefore, state-of-the-art trust and reputation systems combine direct and indirect evidence to estimate trust (Jøsang et al. 2007). SL allows us to combine evidence (opinions) from different sources and compute trust based on both forms of evidence. Let an agent x have evidence $\left(r_y^x, s_y^x\right)$ regarding the trustworthiness of another agent y. Based on this evidence, we can calculate the opinion of x about y's trustworthiness, which we refer to as w_y^x. If x does not have any direct evidence about y (i.e., $r_y^x = 0$ and $s_y^x = 0$), this opinion would be $(0, 0, 1)$ and is of no use to the agent. Similarly, if r_y^x and s_y^x are low, the uncertainty in w_y^x would be high. Therefore, the agent x may obtain other agents' opinions about y's trustworthiness and fuse them into a single, more accurate, opinion. Following the literature (Zhang and Cohen 2006, 2007; Şensoy et al. 2009), we call these agents as *advisers*.

Let us assume that an adviser ϱ provides x with its opinion $w_y^\varrho = \left(b_y^\varrho, d_y^\varrho, u_y^\varrho\right)$ about the trustworthiness of y. The agent x may not directly fuse this opinion with its own, because ϱ may be incompetent in evaluating the trustworthiness of y or a malicious agent trying to mislead x. Therefore, w_y^ϱ is discounted by t_ϱ^x, which is x's opinion about the trustworthiness of ϱ as an adviser (Jøsang 2011). The discounted opinion is referred to as $w_y^{x:\varrho}$ and computed as in equation (5).

$$
\begin{aligned}
w_y^{x:\varrho} &= \left(b_y^{x:\varrho}, d_y^{x:\varrho}, u_y^{x:\varrho}\right), \quad \texttt{where} \\
b_y^{x:\varrho} &= b_y^\varrho \times t_\varrho^x \\
d_y^{x:\varrho} &= d_y^\varrho \times t_\varrho^x \\
u_y^{x:\varrho} &= 1 - b_y^{x:\varrho} - d_y^{x:\varrho} .
\end{aligned}
\tag{5}
$$

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The uncertainty in the discounted opinion increases as the trustworthiness of the adviser decreases. For instance, when $t_{\varrho}^{x} = 0$, the discounted opinion becomes $(0, 0, 1)$ and does not affect the opinion of x about the trustworthiness of y.

Each opinion corresponds to a set of evidence as described before. In this article, we assume that the evidence (i.e., the opinions) provided by each adviser is independent (Etuk, Norman, and Sensoy 2012). This is meaningful when all advisers report opinions based on their own direct interactions; i.e., only direct evidence is used to generate communicated opinions. Based on this assumption, we can easily fuse two or more opinions using the *cumulative fusion* operator $\oplus$ of SL (Jøsang 2011). Let $op(r, s)$ be a function that maps evidences to opinions using equation (2). Then, the fusion of w_y^x with the discounted opinions $\{w_y^0, \ldots, w_y^n\}$ is computed as in equation (6)

$$fuse\left(w_y^x, w_y^0, \ldots, w_y^n\right) = op\left(r\left(w_y^x\right) + \sum_{i=0}^{n} r\left(w_y^i\right), s\left(w_y^x\right) + \sum_{i=0}^{n} s\left(w_y^i\right)\right). \quad (6)$$

This operator converts each discounted opinion into corresponding positive and negative evidence using equation (3), then sums all positive and negative evidences, and last converts the resulting evidence back to the corresponding opinion using equation (2).

In summary, to compute the overall trustworthiness of y, the agent x uses not only direct evidence (i.e., its own opinion) but also the opinions of others (i.e., advisers). Example 1 describes this using a simple scenario. In this approach, it is critical to determine the best degree of discounting (i.e., trustworthiness of advisers) to avoid being misled by malicious or incompetent advisers.

Example 1. Bob wants to asses the trustworthiness of Jeff. He has made only two transactions with Jeff in the past, and their outcomes were quite positive. Therefore, Bob's direct evidence about Jeff is (2, 0). This implies that Bob's opinion is (0.5, 0.0, 0.5), which is very uncertain. To have a more certain opinion, Bob asks Alice for her opinion about Jeff. Alice exposes her opinion as (0.0, 0.8, 0.2), which is mapped to evidence space as (0, 8). Bob fuses his opinion with Alice's opinion based on t_a^b, i.e., Bob's trust in Alice as an adviser.

(i) If $t_a^b = 0$, the fused opinion would be $(0.5, 0.0, 0.5)$.
(ii) If $t_a^b = 1.0$, the resulting evidence about Jeff will be $(2, 8)$, and hence, the fused opinion will be $(0.17, 0.67, 0.16)$.
(iii) If $t_a^b = 0.5$, the discounted opinion of Alice will be $(0.0, 0.4, 0.6)$, which is mapped to $(0, 1.33)$ in evidence space. Then, the resulting evidence about Jeff will be $(2, 1.33)$, and hence, the fused opinion becomes $(0.37, 0.25, 0.38)$.

In the following sections, we first propose an approach for estimating the trustworthiness of advisers. Then, we describe how we can exploit the approach proposed in Section 4 to compute a priori trust in advisers.

5.1. Estimating Trust in Advisers

To estimate the trustworthiness of an adviser, we can use the same model described in Section 2 and formalized in equation (4). This model is based on *evidence* about the trustworthiness of the adviser and a *base rate* (i.e., a priori trust). In this section, we describe what can be counted as evidence for and against the trustworthiness of an adviser.

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To derive some evidence for the trustworthiness of adviser ϱ, agent x can use the opinions of ϱ about their commonly encountered trustees. Let z be a trustee, with which both ϱ and x have had a number of past interactions. Let w_z^ϱ be the opinion of ϱ about the trustworthiness of z and I_z^x be the set of x's direct interactions with z (i.e., transactions). The transactions in I_z^x must occur before w_z^ϱ is provided and must belong to the most recent time window. If the uncertainty in w_z^ϱ is sufficiently low, the truster x uses it to derive some evidence from I_z^x about the trustworthiness of ϱ as an adviser. To set the acceptable upper limit for uncertainty, we use a threshold λ_u, which is set to $1/3$ in our implementation. The higher the upper limit, the more accurate the derived evidence will be. However, as λ_u increases, it may become harder to find sufficiently certain opinions of the advisers to derive evidence.

To be used to derive evidence, the opinion w_z^ϱ should indicate if z is trustworthy or not. In the literature, expectation values of opinions are commonly used for trust decisions (Jøsang et al. 2007), and we follow this practice and use expectation values to derive evidence. For the truster x, w_z^ϱ implies that z is trustworthy if $E\left(w_z^\varrho, a_z^x\right) \geq \gamma$ and z is untrustworthy if $E\left(w_z^\varrho, a_z^x\right) \leq \delta$ where γ and δ are thresholds ($0 \leq \delta < 0.5 < \gamma \leq 1$), and a_z^x is the base rate used by x for the trustworthiness of z. Now, we can use the outcome of each interaction in I_z^x to derive evidence about the trustworthiness of ϱ as an adviser as follows. Given that the outcome of an interaction $i \in I_z^x$ is positive (e.g., a successful or satisfactory transaction),

- i is positive evidence for ϱ if $E\left(w_z^\varrho, a_z^x\right) \geq \gamma$.
- i is negative evidence for ϱ if $E\left(w_z^\varrho, a_z^x\right) \leq \delta$.

Given that the outcome of an interaction $i \in I_z^x$ is negative (e.g., a failed or unsatisfactory transaction),

- i is negative evidence for ϱ if $E\left(w_z^\varrho, a_z^x\right) \geq \gamma$.
- i is positive evidence for ϱ if $E\left(w_z^\varrho, a_z^x\right) \leq \delta$.

Example 2 demonstrates how evidence about trustworthiness of an adviser is calculated using commonly encountered trustees.

Example 2. Bob in Example 1 wants to estimate the trustworthiness of Alice before using her opinion about Jeff. Previously, Bob asked Alice her opinions about the trustworthiness of other agents. Alice provided her opinions for Joe, April, and Jane as (0.1, 0.8, 0.1), (0.2, 0.7, 0.1), and (0.75, 0.2, 0.05), respectively. Bob recently made a number of transactions with Joe and Jane, but he has not interacted with April yet. Bob had four successful (i.e., satisfactory) and 30 unsuccessful (i.e., unsatisfactory) transactions with Joe. Bob had eight successful (i.e., satisfactory) and 10 unsuccessful (i.e., unsatisfactory) transactions with Jane. Bob's opinions about the trustworthiness of Joe and Jane based on the outcomes of these transactions are (0.2, 0.75, 0.05) and (0.4, 0.5, 0.1), respectively. The beta distributions for these opinions and their expectation values are shown in Figure 6, where the base rate for the trustworthiness of Joe and Jane is taken as 0.5. Alice's opinion about Joe indicates that the transactions with Joe mostly fail. Bob's 30 transactions with Joe confirm Alice's opinion while four of Bob's transactions do not. Alice's opinion about Jane indicates that the transactions with Jane mostly succeed. Bob's eight transactions with Jane confirms Alice's opinion while 10 of Bob's transactions do not. Therefore, the derived evidence for Alice's trustworthiness as an adviser is (38, 14). This evidence is derived from Alice's opinion

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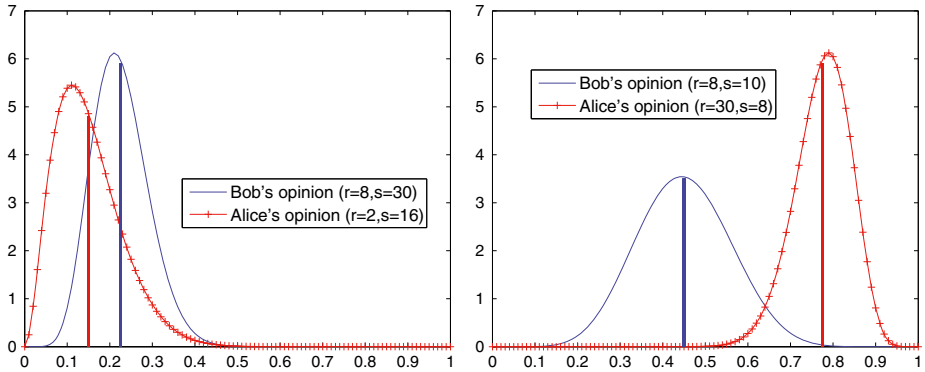

(a) Opinions about Joe (b) Opinions about Jane

FIGURE 6. Beta distributions with their expectation values (given $a = 0.5$) for the opinions in Example 2.

about Joe and Bob's experience. Based on this evidence, Bob's opinion about the trustworthiness of Alice as an adviser is computed as (0.7, 0.26, 0.04). In these calculations, we use $\lambda_u = 1/3, \delta = 1/3$, and $\gamma = 2/3$.

In our model, evidence for and against advisers is derived by an agent based on their opinions about commonly encountered trustees. If the agent and an adviser have few or no commonly encountered trustees, our approach can derive little evidence. In these settings, the agent's opinion about the trustworthiness of the adviser would be mostly or entirely uncertain, and the trust in the adviser would be determined by the a priori trust, as shown in equation (4). Hence, it is critical to determine a priori trust for advisers correctly.

5.2. Estimating A Priori Trust for Advisers

Our approach in Section 4 discovers informative common patterns derived from semantic representation of agents' attributes and relationships. Then, it uses these discovered frequent patterns to estimate a priori trust for agents. This approach is flexible enough to be used for computing a priori trust for advisers as well. Once the a priori trust is determined, it is used while estimating trust in advisers in the lack of evidence about their trustworthiness as information sources. However, the approach in Section 4 builds over semantic representation of attributes and relationships. Hence, to use it, each adviser should be described with its known attributes and relationships.

The attributes and relationships of advisers can be of any type. Therefore, the description of an adviser captures not only the attributes of the adviser but also its social network and even the attributes of other agents and entities in this network. In Figure 7, we provide an example description for an adviser named *Alice*.[1] This approach allows us to capture subtle details about an adviser and its social network. We believe that this is important because existing work on malicious behavior reveals that malicious nodes can be detected by analyzing their social networks (Yu 2011). In Sybil and whitewashing attacks, malicious agents create fake identities to manipulate a system. However, these identities may have some common patterns in their profiles, descriptions, and social networks. If such patterns can be discovered, they can be exploited to estimate a priori trust for advisers.

[1] We assume the existence of some ontologies where the entities in Figure 7(a) are defined semantically and the semantic information is used to extend graph in Figure 7(a) into Figure 7(b).

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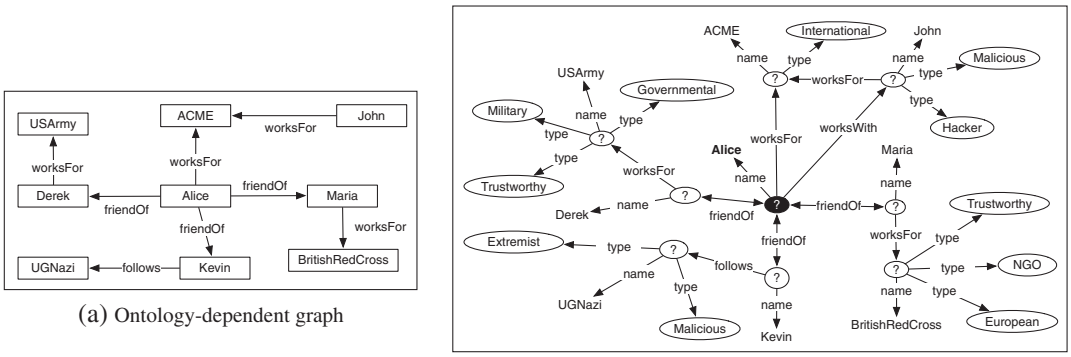

(a) Ontology-dependent graph

(b) Ontology-independent graph

FIGURE 7. Description of the adviser *Alice*. For conciseness, only some of the attributes and relations are shown.

The description of Alice and her social network in Figure 7 presents a number of patterns. Based on four of these patterns, she can be defined as follows: (1) someone working with a malicious entity, (2) a friend of someone following a malicious entity, (3) a friend of someone working for government, and (4) a friend of someone working for a trustworthy entity. Although we can drive these patterns easily from the description of Alice using ontological knowledge, we do not know which of these patterns may be informative about the trustworthiness of Alice as an adviser yet.

To discover the most informative patterns in the descriptions of advisers, we use frequent pattern mining as described in Section 4.1. That is, ontology-independent graphs for known advisers are labeled based on the trustworthiness of these advisers. Then, the labeled graphs are used as a training set to discover significant patterns (i.e., subgraphs) that frequently exist in either trustworthy or untrustworthy advisers. Once the significant patterns are discovered, we use these patterns to train a regression model as described in Section 4.2. The trained model is used to estimate a priori trust for an adviser, given its description as an ontology-independent graph.

6. EVALUATION

In this section, we first evaluate our approach with trustworthy advisers to clearly demonstrate its benefits in estimation of a priori trust for trustees. Then, we extend our evaluation into settings with malicious advisers.

6.1. Malicious Trustees and Honest Advisers

In evaluating our approach, we employed a simulated agent society where a set of truster agents interact with a set of trustee agents over a number of rounds. Each trustee is assigned a performance profile that determines how it will behave. Each profile specifies the mean and standard deviation parameters of a Gaussian distribution from which simulated interaction outcomes will be drawn. A truster considers an interaction's outcome as a *success* if it is greater than a threshold λ; otherwise, it is considered a *failure*. This threshold could vary for each truster, so that different trusters may perceive the same outcome differently. However, for simplicity, we assume that all trusters use the same threshold value. Table 1 uses the values of all parameters used in the simulations, where ψ is the probability that agents

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TABLE 1. Experimental Parameters.

Parameter	Value	Description
$\alpha_- = \alpha_+$	0.1	Support constraints
σ	2	Frequency coefficient
Δ	10	Learning interval
λ	0.5	Success threshold
λ_u	$1/3$	Uncertainty threshold
γ	$2/3$	Trust threshold
δ	$1/3$	Distrust threshold
ψ	[0, 1]	Probability of leaving society
$Life$	500	Simulation time
N_a	100	Number of trustee agents
N_t	20	Number of truster agents
N_i	100	Total number of concept instances
N_{avail}	10	Number of available trustees
N_{adv}	10	Number of advisers

TABLE 2. Profiles and Performance Properties.

Profile ID	Mean	Standard Deviation
P_1	0.9	0.05
P_2	0.6	0.15
P_3	0.4	0.15
P_4	0.3	0.1
P_5	0.0	1

leave and join the society, and Δ is the learning interval that determines when agents should discover new patterns.

In this work, while evaluating if an agent is untrustworthy or trustworthy, the thresholds δ and γ (s.t., $0 \leq \delta < 0.5 < \gamma \leq 1$) are used. The gap between these threshold defines the space between trust and distrust. We use these thresholds to manage this gap. In our evaluation (Table 1), we use $\delta = 1/3$ and $\gamma = 2/3$. Marsh and Dibben (2005) introduced the notion of *untrust* that resides in this gap between trust and distrust. They examined the need for these concepts, highlighting how they interlink and why they can be useful (Marsh and Dibben 2005). While important, detailed discussion of these concepts is out of the scope of this article.

In our experiments, we associate each profile with certain attributes and patterns. Using the proposed approach, the truster tries to learn this association and exploits it to bootstrap trust. Table 2 lists the profiles used in our evaluations. The profile P_1 represents a completely reliable class of agents, while P_4 represents agents who will always perform poorly. Profiles P_2 and P_3 represent unreliable agents who may perform well or poorly, and P_5 represents agents with very wide performance distributions.

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Each trustee y has a set of attributes, each defined as a triple $\langle y, hasRel, i_{k:j} \rangle$ ($hasRel(y, i_{k:j})$ hereafter) where *hasRel* is the object property used to define attributes and $i_{k:j}$ is the jth instance of a leaf concept C_k in the concept hierarchy. The hierarchy of concepts used in our simulations is shown in Figure 8.

We have 20 trusters and 100 trustees in our simulations. Each profile has an equal number of trustees, and each trustee behaves according to one of the five profiles. Each simulation is run for 500 discrete time steps. At each step, each truster selects one trustee to interact with. From a list of 10 randomly selected available trustees in the environment, it evaluates their trustworthiness using a trust model and selects the one with the highest estimated trust value. We compare the following trust models in our experiments.

- *BRS:* Trust is estimated using equation (4) where a is set to 0.5 (Jøsang and Ismail 2002). In this approach, advisers propagate their ratings about trustees. A rating of an adviser ϱ to a trustee y is of the form (r, s), where r is the number of good interactions that ϱ has had with y (i.e., positive evidence) and s is the number of bad interactions that ϱ has had with y (i.e., negative evidence). Ratings from different advisers about the same trustee are combined by simply computing the total number of good interactions and the total number of bad interactions with the trustee as in equation (6). In this approach, ratings from advisers are not discounted before fusion, because it assumes that all advisers are trustworthy. The aggregated evidence is then used to compute the parameters of a beta distribution function. The expectation value of this distribution is taken as the trustworthiness of the trustee.
- *Stereotypical trust (ST):* This is same as *BRS* except that trust is estimated using equation (4) where the base rate a is computed using an M5 regression model. This model is trained using observable attributes (a vector of simple features) of known agents and their trustworthiness (Burnett et al. 2010). Similar to *BRS*, this approach assumes that all advisers are trustworthy (although see Burnett et al. (2013) for an extension to this model that addresses the issue of *bias* in advisers).
- *Stereotypical Trust Assessment through Graph Extraction (STAGE)*.

The truster also obtains a list of 10 advisers from the environment and queries them for evidence (i.e., past interaction outcomes) about trustees. In this section, other trusters serve as advisers, and we assume that they honestly share their evidence about trustees. This assumption is made only so that, in this section, we can focus our evaluations on bootstrapping trust.

To simulate dynamic environments, we introduce the parameter ψ that determines the probability that a trustee will leave the society. When a trustee leaves the society, a new trustee with the same profile joins. In this way, we maintain the balance of profiles and also simulate *whitewashing* behavior.

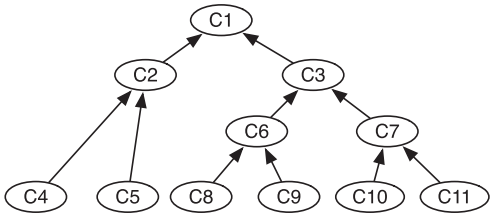

FIGURE 8. Concept hierarchy used in simulations.

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In this section, we evaluate our approach in three settings: (i) performance of trustees is not correlated with their descriptions, (ii) their performance is correlated with their attributes, and (iii) their performance is correlated with the patterns in their descriptions. We repeat each experiment five times and plot the mean for each set of five runs. The significance of the results are assessed using Student's t-test, and where we draw positive conclusions from our results, the p-value is less than 0.05. We use the ratio of successful interactions. For frequent subgraph mining, we use ParMol (Meinl et al. 2006).

6.1.1. No Discriminative Features. In this set of experiments, each trustee is assigned a random set of attributes (i.e., simple features). Hence, there is no correlation between their trustworthiness and features. Figure 9 shows our results for this setting when no trustee leaves the society during the experiments ($\psi = 0.0$). In this setting, there is no significant difference in the performance of the three approaches; they each successfully determine the most trustworthy agents in the environment and rapidly increase their ratio of successful interactions.

Figure 10 shows our results for this setting when ψ is increased to 0.5. In this setting, trust approaches do not have sufficient interaction history to learn the trustworthiness of each trustee. Hence, they have much lower performance compared with the case $\psi = 0$.

We repeated our experiments for different ψ values, the results of which are shown in Figure 11. Our experiments indicate that the proposed approach does not do any worse than the existing approaches we have chosen to compare it with when there is no correlation between features of trustees and their performance. In such settings, statistical trust approaches fail mostly because of a lack of evidence.

6.1.2. Attributes as Discriminative Features. In this set of experiments, we associate the attributes in Figure 12 to profiles as shown in Figure 13. In this way, we set all trustees sharing the same profile to have some common attributes, such that these do not overlap with attributes assigned to other profiles, in addition to their randomly assigned attributes.

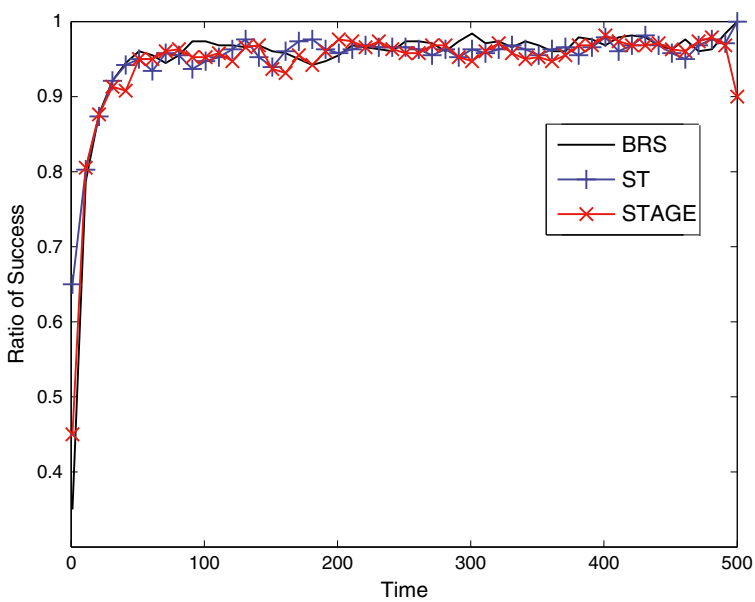

FIGURE 9. Ratio of success when there are no discriminative features ($\psi = 0.0$).

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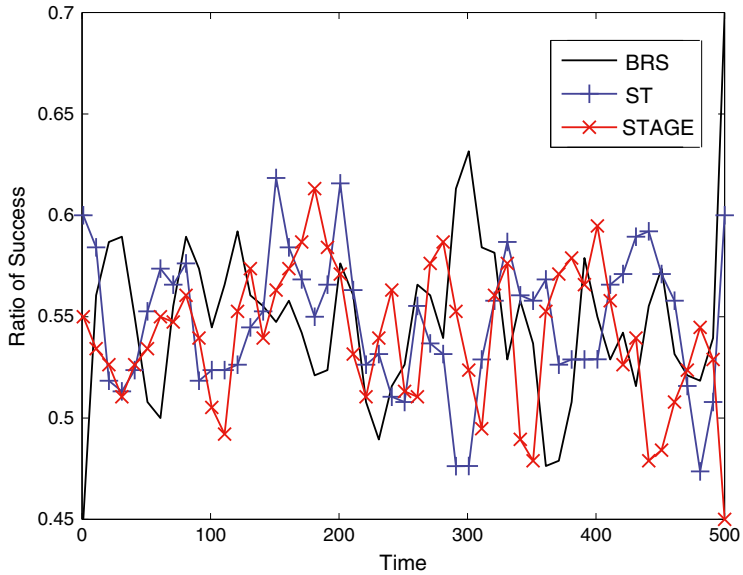

FIGURE 10. Ratio of success when there are no discriminative features ($\psi = 0.5$).

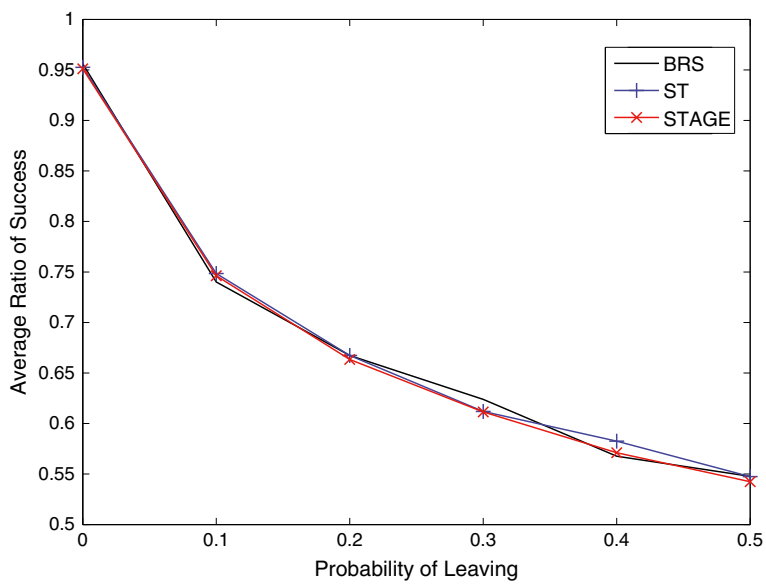

FIGURE 11. Average ratio of success as ψ changes when there are no discriminative features.

Figure 14 shows our results for this setting for different values of ψ. The figure indicates that the performance of *BRS* significantly decreases as the probability of leaving the society increases. *BRS* assumes the same a priori trust for all trustees and build models of the trustworthiness for individual trustees over time. However, as ψ increases, *BRS* does not have enough evidence to compute trust precisely. Unlike *BRS*, *ST* learns stereotypes based on the attributes of trustees. Hence, it exploits the correlation between attributes of trustees and their performance to better estimate a priori trust. As a result, *ST* has a significantly

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a_1	$hasRel(?x, i_{4:1})$
a_2	$hasRel(?x, i_{5:1})$
a_3	$hasRel(?x, i_{8:1})$
a_4	$hasRel(?x, i_{9:1})$
a_5	$hasRel(?x, i_{10:1})$
a_6	$hasRel(?x, i_{11:1})$

FIGURE 12. Attribute definitions.

Profile ID	a_1	a_2	a_3	a_4	a_5	a_6
P_1	X					X
P_2		X		X		
P_3			X	X		
P_4		X	X		X	
P_5		X	X			X

FIGURE 13. Profile descriptions based on attributes.

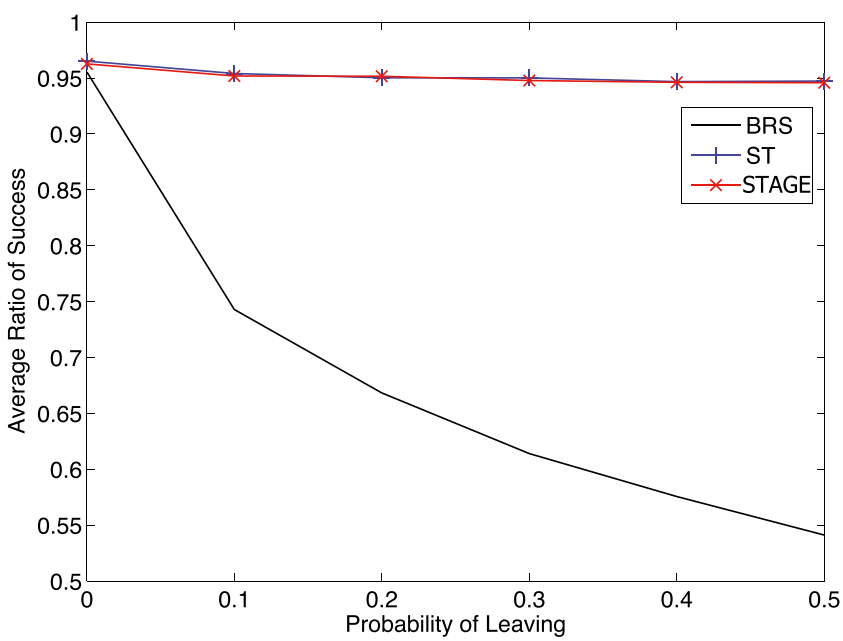

FIGURE 14. Average ratio of success as ψ changes when attributes are discriminative features.

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higher success ratio than *BRS*, and that does not decrease with ψ. *STAGE* successfully discovers the patterns implied by the attributes in Figure 12 and uses these patterns as features to learn a mapping between features and a priori trust. As a result, in this setting, *STAGE* is as successful as *ST* and correctly estimates trustworthiness even at high values of ψ.

6.1.3. Patterns as Discriminative Features. In this set of experiments, we associate patterns from Table 3 with profiles of trustees as shown in Table 4. In this way, we set all trustees sharing the same profile to share some common patterns. Figure 15 indicates that, in this challenging setting, *BRS* and *ST* do not estimate trustworthiness effectively as ψ increases. *ST* fails because it could not find any mapping between attributes of trustees and their trustworthiness. *STAGE*, however, discovers these patterns and exploits them to learn correlations between these features and trustworthiness. The conclusion we draw from these results is that *STAGE* can better estimate the trustworthiness of agents even in highly dynamic environments where more complex patterns can be used to predict behavior.

6.2. Malicious Trustees and Dishonest Advisers

In the previous section, we assumed that all advisers provide honest assessments of trustees. In this section, we let advisers have profiles that determine their trustworthiness when providing evidence about third parties. When an adviser is asked for evidence about

TABLE 3. Patterns and Their Definitions.

p_0	$hasRel(?x, ?a) \wedge type(?a, C_1)$
p_1	$hasRel(?x, ?a) \wedge type(?a, C_2) \wedge hasRel(?x, ?b) \wedge type(?b, C_3)$
p_2	$hasRel(?x, ?a) \wedge type(?a, C_4) \wedge hasRel(?x, ?b) \wedge type(?b, C_6)$
p_3	$hasRel(?x, ?a) \wedge type(?a, C_5) \wedge hasRel(?x, ?b) \wedge type(?b, C_7)$
p_4	$hasRel(?x, ?a) \wedge hasRel(?a, ?b) \wedge hasRel(?b, ?c) \wedge type(?c, C_3)$
p_5	$hasRel(?x, ?a) \wedge hasRel(?a, ?b) \wedge hasRel(?b, ?c) \wedge type(?c, C_6)$
p_6	$hasRel(?x, ?a) \wedge hasRel(?a, ?b) \wedge hasRel(?b, ?c) \wedge type(?c, C_7)$

The variable $?x$ refers to an adviser.

TABLE 4. Profiles Described Using the Patterns in Table 3.

Profile ID	p_1	p_2	p_3	p_4	p_5	p_6
P_1	X	X				
P_2	X		X			
P_3				X		
P_4				X	X	
P_5				X		X

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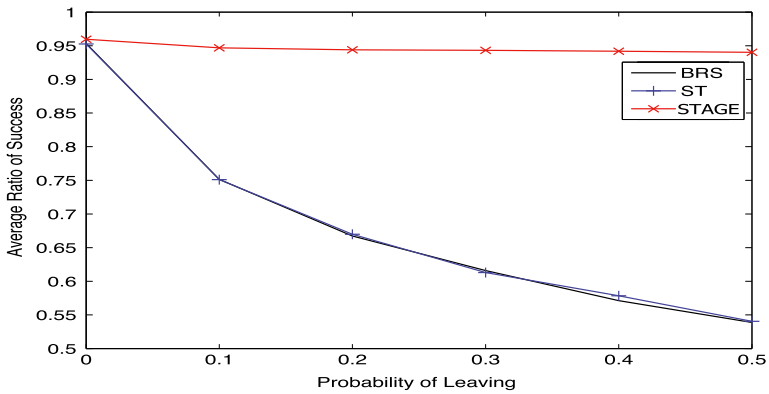

FIGURE 15. Average ratio of success as ψ changes when patterns are discriminative features.

TABLE 5. Profiles of Advisers Based on Their Probability of Lying (P_{lie}).

Adviser profile ID	Mean of P_{lie}	Standard Deviation of P_{lie}
$\bar{P}_1$	0.9	0.05
$\bar{P}_2$	0.1	0.05
$\bar{P}_3$	0.5	0.05

a trustee, it will lie with a probability P_{lie}, which is drawn from a Gaussian distribution based on the profile of the adviser. The mean and variance of P_{lie} for each profile are tabulated in Table 5. $\bar{P}_1$ represents mostly honest advisers with a low P_{lie}, $\bar{P}_2$ represents mostly dishonest advisers with a high P_{lie}, and $\bar{P}_3$ represents advisers whose probabilities of providing honest and dishonest assessments of trustees are equally likely.

Each adviser is assigned only one profile, and each profile is assigned to an equal number of advisers. Lying advisers swap positive and negative evidence before sharing. That is, given that an adviser has r positive and s negative experiences with a trustee, the adviser shares (s, r) as evidence instead of (r, s) when it lies about the trustee. In our simulations, advisers and trustees leave the society with the same probability ψ. When a trustee or adviser leaves the society, a new one with the same profile joins to keep the population stable. As also mentioned in the previous section, ψ allows us to simulate *whitewashing* behavior of advisers and trustees.

In this setting, it is not fair to compare our approach with *BRS* and *ST* because these approaches assume that all advisers are honest. Therefore, for comparison, we use two well-known trust approaches that are proposed for settings with dishonest advisers. These approaches are briefly described later. In the implementation of these approaches, we use the default parameter values given by Whitby, Jøsang, and Indulska (2005) and Teacy et al. (2006).

- *BRS^+:* Whitby et al. (2005) extended *BRS* to filter out unfair ratings (i.e., fake evidence) provided by advisers. This approach filters out those ratings that do not comply with the significant majority of the ratings by using an *iterated filtering approach*. Hence, this

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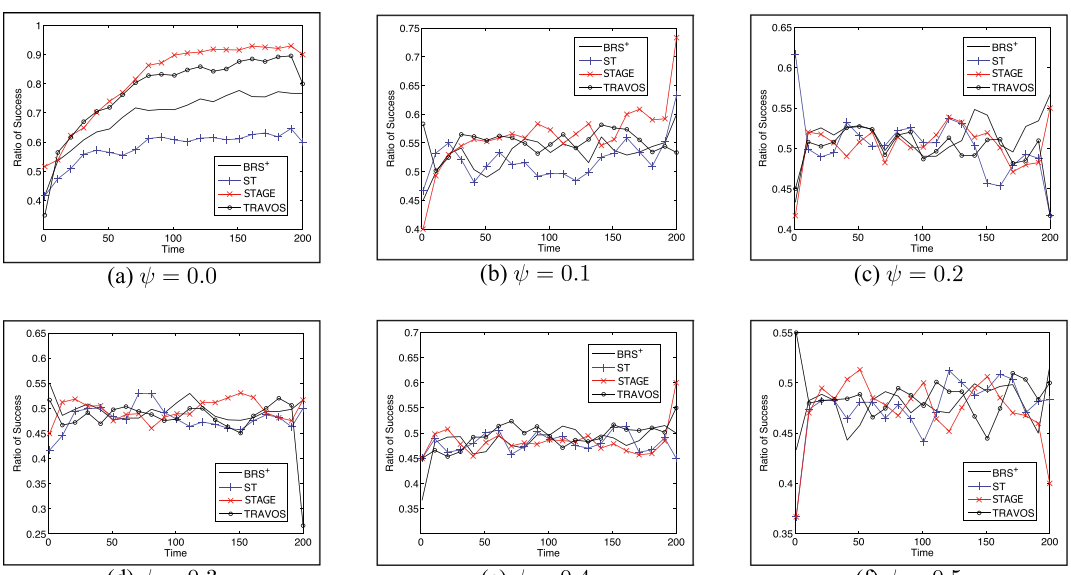
approach assumes that the significant majority of advisers honestly share their ratings; liars are in the minority.

- *TRAVOS:* This approach is proposed by Teacy et al. (2006). Similar to BRS^+, it uses beta probability density functions to compute trust in agents. The main difference between BRS^+ and TRAVOS is the way they filter unfair ratings. While BRS^+ uses the majority of ratings to filter out unfair ratings from advisers, TRAVOS uses the personal observations of trusters about trustees to detect and filter out unfair ratings from malicious advisers. Hence, unlike BRS^+, TRAVOS does not assume that the majority of ratings are fair.

We have extended the ontology used in Section 6.1 with new concepts and roles. Then, based on these concepts and roles, we have defined descriptions (as well as social networks) of trustees and advisers. In the rest of this section, we analyze the performance of our approach in different settings.

6.2.1. No Discriminative Features. In this set of experiments, we randomly create descriptions of trustees and advisers. Therefore, there is no correlation between their trustworthiness and their features. That is, trustees and advisers are randomly assigned to profiles in Tables 2 and 5, respectively. These profiles determine how trustworthy trustees and advisers would be during the experiments. We ran our experiments for 200 epochs for different ψ values and plot our results in Figure 16. When $\psi = 0.0$, advisers and trustees do not leave the society during the experiments. In this setting, as seen in Figure 16(a), *STAGE* learns over time the most trustworthy advisers and trustees. As a result, the ratio of success for *STAGE* approaches and exceeds 0.9 during the experiments. The performance of *TRAVOS* is similar to that of *STAGE*. In this setting, there are no patterns or features of trustees and advisers that may indicate their trustworthiness. Therefore, *STAGE* cannot discover and exploit such patterns for bootstrapping trust. However, because trustees and advisers do



(a) $\psi = 0.0$ (b) $\psi = 0.1$ (c) $\psi = 0.2$

(d) $\psi = 0.3$ (e) $\psi = 0.4$ (f) $\psi = 0.5$

FIGURE 16. Ratio of success when there are no discriminative features.

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not leave, both *STAGE* and *TRAVOS* have an opportunity to model trust over time. Similarly, BRS^+ models trust over time, but its performance is inferior to that of *STAGE* and *TRAVOS*. This is due to the fact that the assumption that a significant majority of advisers are trustworthy does not always hold in our experiments. Among the compared approaches, *ST* is the one with the lowest performance; this is intuitive because *ST* does not have any mechanism for handling untrustworthy advisers.

Our results for different ψ values are shown in Figure 16(b)–(f). As the trustees and advisers leave and join the society ($\psi > 0.0$) in our experiments, the performances of all approaches decrease significantly and become similarly low around 0.5. The change in average ratio of success as ψ varies is demonstrated in Figure 17. The sharp decrease in the performances of *STAGE*, *TRAVOS*, and BRS^+ can be explained by the inability to acquire evidence about the trustees and advisers. *TRAVOS* and BRS^+ require direct or indirect evidence about advisers and trustors to model trust. However, as trustees and advisers frequently leave and join the society, these approaches do not have the opportunity to accumulate evidence about them. Similar to the other approaches, *STAGE* suffers from a lack of evidence, because it cannot find any discriminative features to bootstrap trust assessments.

6.2.2. Patterns as Discriminative Features. In this set of experiments, we associate the patterns of Tables 3 and 6 with the profiles of trustees and advisers as shown in Tables 4 and 7, respectively. Hence, we make all trustees or advisers with the same profile have some common patterns in their descriptions and social networks.

Figure 18 shows the results of our experiments in this setting for different values of ψ. When $\psi = 0.0$, as seen in Figure 18(a), *STAGE* outperforms *TRAVOS*, BRS^+, and *ST*. The performances of the competing approaches improve over time during the experiments, as they have more evidence about trustees and advisers. However, when $\psi > 0$, the approaches other than *STAGE* could not model trust correctly, as clearly shown in Figure 18(b)–(f). The failure of these approaches is due to the lack of evidence about the advisers and trustees, as explained in the previous section. Unlike other approaches, *STAGE* continues to perform

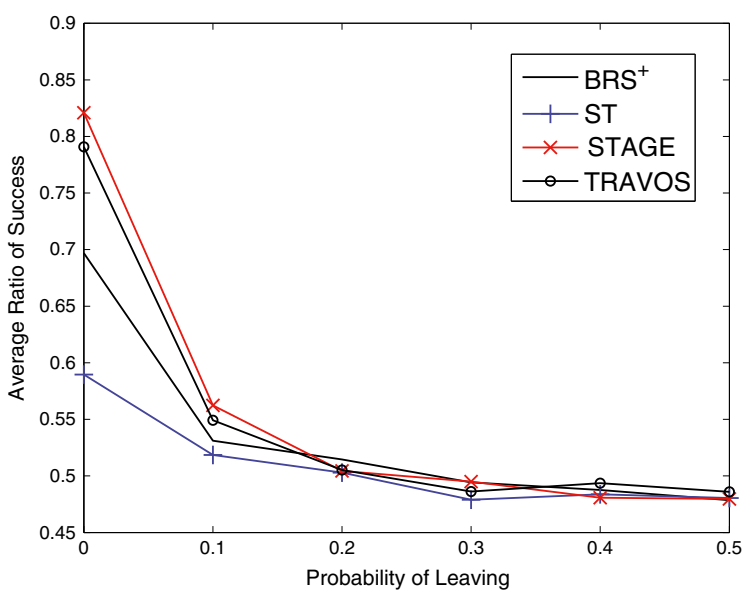

FIGURE 17. Average ratio of success as ψ changes when there are no discriminative features.

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TABLE 6. Patterns for Advisers and Their Definitions.

$\bar{p}_1$	$frindOf(?x, ?y) \wedge worksFor(?y, ?z) \wedge type(?z, Asian)$
$\bar{p}_2$	$frindOf(?x, ?y) \wedge worksFor(?y, ?z) \wedge type(?z, European)$
$\bar{p}_3$	$worksWith(?x, ?a) \wedge type(?a, Malicious)$
$\bar{p}_4$	$follows(?x, ?a) \wedge type(?a, Malicious)$
$\bar{p}_5$	$frindOf(?x, ?y) \wedge worksFor(?y, ?z) \wedge type(?z, Governmental)$
$\bar{p}_6$	$frindOf(?x, ?y) \wedge worksFor(?y, ?z) \wedge type(?z, Trustworthy)$
$\bar{p}_7$	$hasEmail(?x, ?e) \wedge type(?e, Invalid)$

The variable $?x$ refers to an adviser.

TABLE 7. Adviser Profiles Described Using Patterns of Table 6.

Adviser profile ID	$\bar{p}_1$	$\bar{p}_2$	$\bar{p}_3$	$\bar{p}_4$	$\bar{p}_5$	$\bar{p}_6$	
$\bar{p}_7$							
$\bar{P}_1$			X	X			X
$\bar{P}_2$	X	X	X				
$\bar{P}_3$			X		X	X	

very well as ψ increases. In this setting, although the amount of direct and indirect evidence decreases when $\psi > 0.0$, we observe that *STAGE* is able to discover some discriminative features in the descriptions and social networks of advisers and trustees and then these discovered features are used to successfully bootstrap trust in the lack of evidence. Figure 19 demonstrates how the average ratio of success changes for the competing approaches as ψ varies.

7. RELATED AND FUTURE WORK

In this section, we discuss our work with references to the related work and draw directions for future research.

7.1. Related Work

There are a number of models for computing trust and reputation in multiagent systems. Jøsang and Ismail (2002) proposed the BRS. It estimates the reputations of agents using the beta probability density functions. Huynh, Jennings, and Shadbolt (2004) proposed FIRE, an integrated trust and reputation model that consists of four modules: interaction trust, witness reputation, role-based trust, and certified reputation. The interaction trust module

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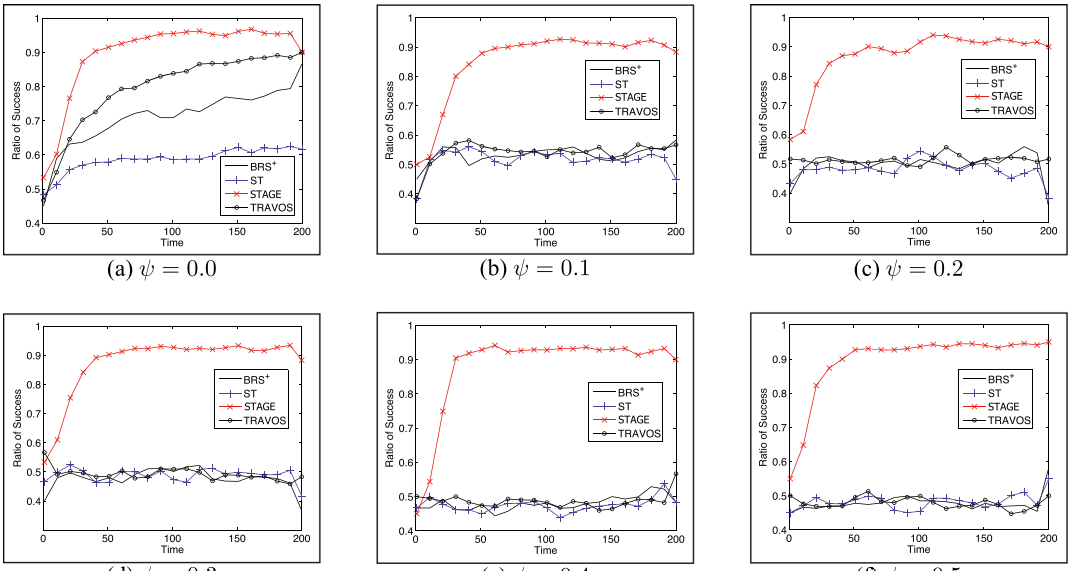

(a) $\psi = 0.0$ (b) $\psi = 0.1$ (c) $\psi = 0.2$

(d) $\psi = 0.3$ (e) $\psi = 0.4$ (f) $\psi = 0.5$

FIGURE 18. Ratio of success when patterns are discriminative features.

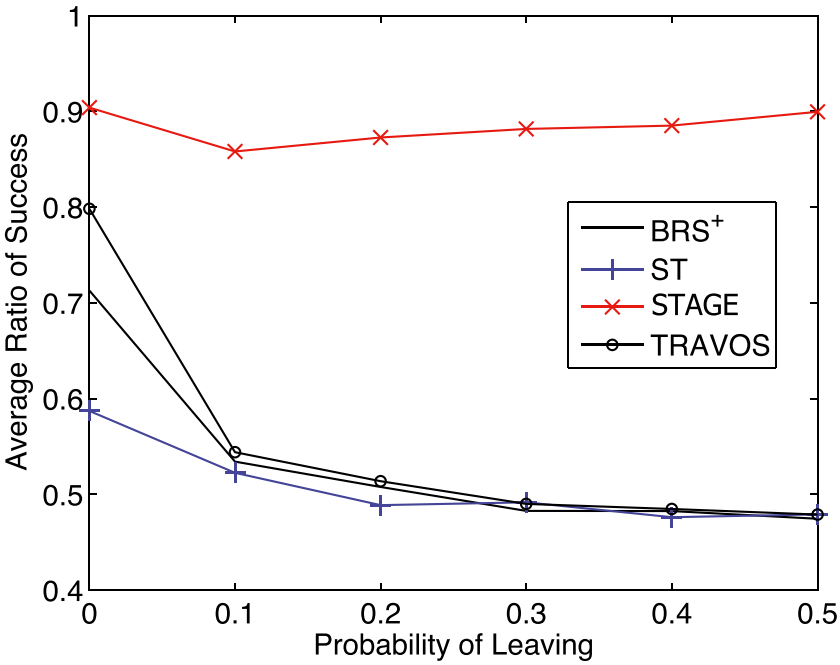

FIGURE 19. Average ratio of success as ψ changes when patterns are discriminative features.

uses only direct evidence while the witness reputation module is based on ratings (i.e., indirect evidence). Unfortunately, these approaches do not have any mechanism for handling unfair ratings (i.e., deceptive indirect evidence) as we do in this work.

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Whitby et al. (2005) extended *BRS* to handle unfair ratings from malicious advisers using a majority-based algorithm. TRAVOS is proposed by Teacy et al. (2006). Similar to BRS, it uses beta probability density functions to compute agents' trust on others. It uses reported ratings about commonly encountered trustees to estimate trust in advisers. Yu and Singh (2003) proposed a trust approach that handles unfair ratings using a version of weighted majority algorithm. In their algorithm, weights are assigned to advisers. These weights are initiated as 1.0, and they can be considered as the trustworthiness of the corresponding advisers. The algorithm makes predictions about trustees based on the weighted sum of ratings provided by those advisers. Yu and Singh proposed to tune the weights of the advisers after an unsuccessful prediction so that the weights assigned to the unsuccessful advisers are decreased. They assume that the ratings of dishonest advisers may conflict with the observations of the trusters receiving these ratings. By decreasing the weights of these advisers over time, unfair ratings are filtered. Zhang and Cohen (2006, 2008) proposed the personalized approach that measures the trustworthiness of an adviser using two metrics: (1) private reputation calculated by comparing the advisers ratings with the personal observations of trusters and (2) public reputation estimated by comparing the advisers ratings and other advisers ratings for all trustees. Noorian, Marsh, and Fleming (2011) proposed a two-layer filtering approach that cognitively elicits the behavioral characteristics of the participating advisers. In the first layer, some advisers are filtered according to the similarity between their ratings and personal observations of trusters. In the second layer, the behavioral models of advisers are exploited to measure their credibility. Regan et al. (2006) proposed BLADE to cope with subjectivity and deception in e-marketplaces. BLADE uses a Bayesian approach to model trustee properties and adviser evaluation functions as dynamic random variables. Then, trusters can progressively learn a probabilistic model that calibrates the interpretation of trustee evaluations provided by advisers.

Teacy et al. (2012) proposed HABIT, a hierarchical Bayesian trust model based on direct and third-party information. Through simulations, it is shown that HABIT is robust in environments in which third-party information is malicious, noisy, or otherwise inaccurate. Unlike similar approaches, HABIT is based exclusively on principled statistical techniques instead of heuristics with little theoretical foundation. It can handle multiple discrete or continuous aspects of agent behavior while allowing different representations of behavior. Furthermore, HABIT can improve trust assessments by using any observed correlation between the behavior of similar trustees or information sources. Through experiments on real and synthetic data sets, Teacy et al. show that HABIT significantly outperforms BLADE in challenging settings. Teacy et al. argue that HABIT tackles the whitewashing problem pragmatically. However, this is purely through behavior recognition of "new" agents that have whitewashed themselves; thus, it is not different from any other trust assessment mechanism in this regard. It might be more effective, but it needs some direct or indirect evidence to classify a new agent; it does not use other (nonbehavioral) factors to classify new agents.

Khosravifar et al. (2012) propose a comprehensive multifactor trust framework. Their framework considers and analyzes multiple factors impacting the trust assessments. Similar to other trust methods, this framework first considers direct interactions among agents. This approach also collects information from consulting agents (i.e., advisers) and evaluates the trustworthiness of these agents by comparing the provided information with direct interaction outcomes. This framework takes into account five important factors: (i) trustworthiness of advisers, (ii) indirect information from advisers, (iii) the number of interactions between advisers and trustee agents, (iv) the timely relevance of the information provided by advisers, and (v) the confidence of advisers. Khosravifar et al. also compared

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their framework with the well-known approaches (e.g., BRS and TRAVOS) through extensive simulations.

Most of the approaches mentioned earlier use direct and indirect past interactions with the other agents to compute trust. However, when a new agent enters to a society, there is no direct or indirect evidence about the agent. Hence, bootstrapping trust becomes a challenge in open and dynamic multiagent systems. To address this challenge, Liu et al. (2009) proposed agents to form stereotypes using their previous transactions with others. In their approach, a stereotype contains certain observable attributes of agents and an expected outcome of the transaction. Similarly, Burnett et al. (2010, 2013) proposed to use stereotyping to bootstrap trust evaluations based on SL. Their approach allows agents to generalize their experience with known agents as stereotypes and apply these when evaluating new and unknown agents. Stereotypes are learned with standard *M5 regression tree* algorithm using a training set composed of observable attributes of known agents and their trustworthiness. Fang et al. (2012) proposed a novel generalized ST model that exploits domain information together with fuzzy membership functions to learn generalized stereotypes. This approach allows stereotypes to capture settings which have not been seen before. However, the learned stereotypes are based on simple and predetermined attributes, as in the other stereotyping approaches. On the other hand, our approach discovers discriminative complex features such as patterns in the description and social networks of agents. Then, these dynamic features are used to learn stereotypes for their trustworthiness. Furthermore, our approach does not assume that all advisers are trustworthy; instead, it discovers distinctive features of advisers as well and learns ST models for them.

The approaches described so far aim to detect and eliminate deceptive third-party information (e.g., opinions, ratings, and so on). On the other hand, there are also other approaches that aim to design mechanisms that promote incentives to provide truthful information and eliminate incentives to lie. These mechanisms give agents (i.e., advisers) some incentives such as digital currencies or credits so that truthful reporting maximizes their expected revenue. Hence, they discourage agents from providing misleading and deceptive information (Jurca and Faltings 2007). While designing these incentive mechanism, usually game-theoretical approaches are used where it is assumed that agents are rational; i.e., they choose actions to maximize their overall profit. Jurca and Faltings (2003, 2007, 2009) proposed payment schemes that create incentives for agents to provide truthful information while reporting their ratings (i.e., feedbacks) about other agents. In their approach, a set of brokers are organized to buy and sell ratings from agents.They devise a payment scheme that pays a submitted rating if it has the same value as randomly chosen feedbacks from other agents. Jurca and Faltings showed that reporting ratings truthfully is the optimum strategy for the agents. Similarly, Miller, Resnick, and Zeckhauser (2005) propose another approach that also creates incentives for agents to share their ratings truthfully. This approach uses a central processing store to reward agents with a suitable payment based on the correlation of their ratings with those of others. However, these approaches may have difficulty when the majority of the agents lie about their ratings or agents' ratings differ because of the factors such as subjectivity, instead of malicious behavior.

Our approach can exploit various kinds of information about agents to learn and estimate their trustworthiness. Our approach is flexible enough to accommodate additional information about the agent when available. This includes not only personal attributes but also social relationships, affiliations, and even more. The source of such knowledge about an agent can be personal observations, other agents that are knowledgeable about the agent, or the Web. Some social networking sites provide information about their members' relationships and affiliations in addition to some demographic information about

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them. Moreover, recent initiatives such as the Friend of a Friend project[2] and the Linked Data[3] provide very valuable information about individuals and institutions. This information is usually publicly available on the Web and can be exploited by our approach with some effort.

Our approach is based on graph mining that is decidable but computationally expensive. Exact algorithms for frequent subgraph mining are NP while there are also some polynomial-time heuristics (Chi et al. 2004). Therefore, as the size and the number of graphs increases, the computational cost of the subgraph mining may increase significantly if exact algorithms are used. While it is out of the scope of this article, here, we shortly discuss these approaches that can be used to alleviate the computational burden of graph mining. The first approach is to use highly distributed graph mining algorithms and heuristics (Cai, Logothetis, and Siganos 2012). The second approach is to carefully reduce the size and number of the graphs before inputting them into graph mining algorithms. To reduce the size of graphs (i.e., agent profiles), we can limit the types of relationships and attributes used in these graphs. Similarly, we can reduce the size of domain ontologies by removing unnecessary concepts and properties. The third approach is to use graph summarization techniques (Fokoue et al. 2012) to generate much smaller representative graphs; then, these representative graphs are used during graph mining instead of the original ones. Hence, the computational load of graph mining can be significantly reduced.

The key contribution of this work is an approach that exploits advanced knowledge representation formalisms and graph-mining techniques to discover and use informative patterns for bootstrapping trust in the lack of evidence. We linked this approach to a particular trust-modeling framework that is based on beta distributions. The adopted framework is well accepted and highly utilized in the statistical trust literature (Jøsang and Ismail 2002; Teacy et al. 2006; Jøsang et al. 2007; Zhang and Cohen 2013). The bootstrapping mechanism that represents our main contribution does not, however, depend on this choice of trust mechanism. The use of graphical patterns to bootstrap trust assessments where there is little direct evidence could be employed in conjunction with a wide range of trust models. This includes non-Bayesian models such as the cognitive approach taken by Castelfranchi and Falcone (2010) where the output of our mechanism can be treated as an additional belief about the subject.

7.2. Future Work

Simulations provide a controlled environment to evaluate performances of trust approaches in multiagent systems under various challenging settings. Simulations also provide these approaches an interactive environment that could not be found in publicly available real-life data sets. This is why all of the existing approaches mentioned earlier and our approach in this article are evaluated through simulations. As a future work, we want to design a *game with a purpose* to test our approach in a controlled interactive environment with human users.

The main contribution of this work is the mechanisms for bootstrapping trust in the lack of evidence. It is important to point out that these mechanisms can also be used to augment other trust approaches. In future, we want to demonstrate how the techniques proposed in this work can be used to enhance the capabilities of the existing approaches.

The proposed approach dynamically learns new patterns and rules out existing ones if their significance decreases over time. Therefore, we believe that our approach can

[2] http://www.foaf-project.org/.

[3] http://linkeddata.org/.

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handle the situations where agents strategically change their profiles or collude in sophisticated ways for whitewashing. For example, if most malicious agents use a free Web hosting service for their Web sites, we may discover that agents using free hosting are less trustworthy and the ones using paid services are more trustworthy. If the malicious agents start to use paid hosting services, this modification in their profile involves a cost. If many of the malicious agents are willing to pay this cost, after a while, the proposed approach discovers that using paid Web hosting services is not significant for trust estimation any more. That is, this pattern does not discriminate trustworthy agents from untrustworthy ones any more. Similarly, malicious agents may collude to promote some patterns as trustworthy by being honest when these patterns are in their profiles. However, when they start to exploit these trustworthy patterns to deceive the system, our approach will discover that these patterns are not trustworthy any more because they appear in the profiles of deceiving agents. In future, we want to analyze these settings theoretically and experimentally.

The proposed approach is based on profiles of agents. However, agent profiles may be incomplete, and some important attributes of agents may not be known. The effect of missing attributes in agent profiles is another interesting and important research direction.

8. CONCLUSIONS

Humans build stereotypes to evaluate others in the absence of personal familiarity (Hurst 1992). As a working human decision-making mechanism under uncertainty, stereotyping is built upon discovering and exploiting the correlation between observable features of others and their behavior. Inspired from this real-life phenomenon, we argue in this article that agents with similar behavior may also share some common patterns in their descriptions or relationships. To capture the subtle details about the agents and their relationships, we use state-of-the-art knowledge representation and reasoning formalisms based on Semantic Web ontologies (Antoniou and van Harmelen 2004). We propose to discover these significant patterns in trustworthy and untrustworthy agents, and exploit them to bootstrap trust in dynamic and uncertain environments. To discover patterns in semantic descriptions of agents, we have extended existing frequent subgraph mining techniques. Our approach is flexible enough to model and bootstrap trust both for trustees and advisers. Through extensive empirical evaluations, we show that the proposed approach is at least as good as the existing approaches in all settings. However, it significantly outperforms existing approaches when agents (i.e., trustees and advisers) with similar behavior share some common patterns in their descriptions or social networks.

Our approach allows effective handling of untrustworthy advisers where complex patterns of attributes and social connections can offer evidence regarding their tendency to give false assessments of third parties. We neither rely on the majority being truthful as in BRS^+, nor is this vulnerable to whitewashing as with TRAVOS. This complements models such as TRAVOS and BLADE as we propose a means to exploit additional, contextual evidence where available to improve trust assessments.

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